

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			Steady state (1) & Big bang (1) ticked 3 boxes ticked (-1) 4 boxes ticked (0)	2			2		
	(b)			Radio – micro(waves) – IR – visible / light – UV – X-rays – gamma (rays) Accept gamma symbol Any 2 × (1)	2			2		
	(c)			helium ticked 2 ticks = 0 marks			1	1		
	(d)	(i)		Mars		1		1		
		(ii)		2 Accept: Earth and Mars		1		1		
		(iii)		Jupiter		1		1		
		(iv)		(No) because it is further away than Neptune. OR (No) otherwise its orbit would be between 228 and 779. OR (No) if orbit between Mars and Jupiter it would be less than 10 180 OR Mars and Jupiter are only 228 and 779 from the Sun, so Eris can't be found between then			1	1		
	(e)			Middle orbit ticked	1			1		
				Question 4 total	5	3	2	10		